Project Design Phase-I

Proposed Solution Template

Date	5 March 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Project - xxx
Maximum Marks	2 Marks

Proposed Solution Template:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Electricity consumption is increasing rapidly, but many regions do not properly analyze their energy usage patterns. Because of this, it becomes difficult to identify electricity wastage, high-consumption areas, and inefficient distribution. Without proper analysis, energy providers cannot make effective decisions for power management and conservation..
2.	Idea / Solution description	The solution is an Electricity Consumption Analysis system that uses data analytics and visualization to study electricity usage patterns. The system processes consumption data and presents insights through charts and dashboards, helping users easily understand trends and compare electricity usage across regions and time periods.
3.	Novelty / Uniqueness	The uniqueness of this project is the use of data visualization to convert large electricity datasets into clear and interactive insights. Instead of raw numbers, the system provides

		visual analysis that makes it easier to identify patterns, trends, and areas of high energy consumption.
4.	Social Impact / Customer Satisfaction	This solution helps governments and energy providers understand electricity usage better and reduce wastage. Efficient energy management can lead to better power distribution, lower electricity costs, and more reliable electricity supply for consumers.
5.	Business Model (Revenue Model)	The system can be used by energy companies, government agencies, and research organizations. Revenue can be generated through analytics services, subscription-based dashboards, or consulting services for energy consumption analysis.
6.	Scalability of the Solution	The solution is scalable because it can analyze electricity data from multiple regions and large datasets. It can be expanded to include real-time monitoring, smart grid integration, and advanced analytics for better energy planning in the future.